

PACKAGE INSERT

ZOBREX

Brand or Product Name

Zobrex Capsule 200 mg

Name and Strength of Active Substance

Celecoxib 200 mg

Product Description

White, No.2 hard gelatin capsule, with " MacroPhar " overprinted in black colour on capsule containing white granular powder.

Pharmacodynamics/ Pharmacokinetics

Pharmacodynamics

Pharmacotherapeutic group: Non-steroidal anti-inflammatory and antirheumatic drugs, NSAIDs, Coxibs.

ATC code: M01AH01.

Celecoxib is an oral, selective, cyclooxygenase-2 (COX-2) inhibitor within the clinical dose range (200-400 mg daily).

Cyclooxygenase is responsible for generation of prostaglandins. Two isoforms, COX-1 and COX-2, have been identified. COX-2 is the isoform of the enzyme that has been shown to be induced by pro-inflammatory stimuli and has been postulated to be primarily responsible for the synthesis of prostanoid mediators of pain, inflammation, and fever. COX-2 is also involved in ovulation, implantation and closure of the ductus arteriosus, regulation of renal function, and central nervous system functions (fever induction, pain perception and cognitive function). It may also play a role in ulcer healing. COX-2 has been identified in tissue around gastric ulcers in man but its relevance to ulcer healing has not been established.

The difference in antiplatelet activity between some COX-1 inhibiting NSAIDs and COX-2 selective inhibitors may be of clinical significance in patients at risk of thrombo-embolic reactions. COX-2 selective inhibitors reduce the formation of systemic (and therefore possibly endothelial) prostacyclin without affecting platelet thromboxane.

Celecoxib is a diaryl-substituted pyrazole, chemically similar to other non-arylamine sulfonamides (e.g. thiazides, furosemide) but differs from arylamine sulfonamides (e.g. sulfamethoxazole and other sulfonamide antibiotics).

Pharmacokinetics

Celecoxib is well absorbed reaching peak plasma concentrations after approximately 2-3 hours. Dosing with food (high fat meal) delays absorption by about 1 hour.

Celecoxib is mainly eliminated by metabolism. Less than 1% of the dose is excreted unchanged in urine. The inter-patient variability in the exposure of celecoxib is about 10-fold. Celecoxib exhibits dose- and time-independent pharmacokinetics in the therapeutic dose range. Plasma protein binding is about 97% at therapeutic plasma concentrations and the drug is not preferentially bound to erythrocytes. Elimination half-life is 8-12 hours. Steady state plasma concentrations are reached within 5 days of treatment. Pharmacological activity resides in the parent drug. The main metabolites found in the circulation have no detectable COX-1 or COX-2 activity.

Celecoxib metabolism is primarily mediated via cytochrome P450 2C9. Three metabolites, inactive as COX-1 or COX-2 inhibitors, have been identified in human plasma i.e., a primary alcohol, the corresponding carboxylic acid and its glucuronide conjugate.

Cytochrome P450 2C9 activity is reduced in individuals with genetic polymorphisms that lead to reduced enzyme activity, such as those homozygous for the CYP2C9*3 polymorphism.

Patients who are known, or suspected to be CYP2C9 poor metabolizers based on previous history/experience with other CYP2C9 substrates should be administered celecoxib with caution (see Recommended Dosage).

No clinically significant differences were found in PK parameters of celecoxib between elderly African-Americans and Caucasians.

The plasma concentration of celecoxib is approximately 100% increased in elderly women (>65 years).

Patients with severe hepatic impairment (serum albumin <25 g/l) have not been studied and celecoxib is contraindicated in this patient group.

There is little experience of celecoxib in renal impairment. The pharmacokinetics of celecoxib has not been studied in patients with renal impairment but is unlikely to be markedly changed in these patients. Thus caution is advised when treating patients with renal impairment. Severe renal impairment is contraindicated.

Indication

For the management of acute pain in adults and for the treatment of primary dysmenorrhoea.

Relief of the acute and chronic pain and inflammation of rheumatoid arthritis and osteoarthritis.

Relief of signs and symptoms of ankylosing spondylitis.

For the management of low back pain.

Recommended Dosage

Celecoxib capsules can be taken with or without food.

Given the association between cardiovascular risk and exposure to COX-2 inhibitors, doctors are advised to use the lowest effective dose for the shortest possible duration of treatment.

Adults

Symptomatic Treatment of Osteoarthritis (OA): The recommended dose of celecoxib is 200 mg administered as a single dose or as 100 mg twice per day.

Symptomatic Treatment of Rheumatoid Arthritis (RA): The recommended dose of celecoxib is 100 mg or 200 mg twice per day.

Ankylosing Spondylitis (AS): The recommended dose of celecoxib is 200 mg administered as a single dose or 100 mg twice per day. Some patients may benefit from a total daily dose of 400 mg.

Management of Acute Pain: The recommended dose of celecoxib is 400 mg, initially, followed by an additional 200 mg dose, if needed on the first day. On subsequent days, the recommended dose is 200 mg twice daily, as needed.

Treatment of Primary Dysmenorrhea: The recommended dose of celecoxib is 400 mg, initially, followed by an additional 200 mg dose, if needed on the first day. On subsequent days, the recommended dose is 200 mg twice daily, as needed.

Low Back Pain (LBP): Usual dosage for adults is 100 mg of celecoxib orally twice daily, morning and evening after meal, or 200 mg once daily (100 mg and 200 mg only)

Elderly: No dosage adjustment is generally necessary. However, for elderly patients with a lower than average body weight (<50 kg), it is advisable to initiate therapy at the lowest recommended dose.

Hepatic impairment: No dosage adjustment is necessary in patients with mild hepatic impairment (Child-Pugh Class A). Introduce celecoxib at half the recommended dose in arthritis or pain patients with moderate hepatic impairment (Child-Pugh Class B).

Patients with severe hepatic impairment (Child-Pugh Class C) have not been studied (see Warnings and Precautions – Hepatic Effects).

Renal impairment: No dosage adjustment is necessary in patients with mild or moderate renal impairment. There is no clinical experience in patients with severe renal impairment (see Warnings and Precautions – Renal Effects).

Children: Celecoxib is not indicated for use in children.

CYP2C9 Poor Metabolizers: Patients who are known, or suspected to be CYP2C9 poor metabolizers based on genotyping or previous history/experience with other CYP2C9 substrates should be administered celecoxib with caution as the risk of dose-dependent adverse effects is increased. Consider reducing the dose to half the lowest recommended dose (see Pharmacokinetic – Metabolism).

Route of Administration:

Oral use

Contraindications

History of hypersensitivity to the active substance or to any of the excipients.

Known hypersensitivity to sulphonamides.

Active peptic ulceration or gastrointestinal (GI) bleeding.

Patients who have experienced asthma, acute rhinitis, nasal polyps, angioneurotic oedema, urticaria or other allergic-type reactions after taking acetylsalicylic acid or NSAIDs including COX-2 (cyclooxygenase-2) inhibitors.

In pregnancy and in women of childbearing potential unless using an effective method of contraception (See Statement on usage during Pregnancy and Lactation). The potential for human risk in pregnancy is unknown, but cannot be excluded.

Breast feeding (See Statement on usage during Pregnancy and Lactation).

Severe hepatic dysfunction (serum albumin <25 g/l or Child-Pugh score ≥ 10).

Patients with estimated creatinine clearance <30 ml/min.

Inflammatory bowel disease.

Congestive heart failure (NYHA II-IV).

Established ischaemic heart disease, peripheral arterial disease and/or cerebrovascular disease.

Contraindication for patients who have increased risk of cardiovascular disease (ischemic heart disease and stroke).

Warning and Precautions

Upper gastrointestinal complications (perforations, ulcers or bleedings (PUBs)), some of them resulting in fatal outcome, have occurred in patients treated with celecoxib. Caution is advised with treatment of patients most at risk of developing a gastrointestinal complication with NSAIDs; the elderly, patients using any other NSAID or acetylsalicylic acid concomitantly or patients with a prior history of gastrointestinal disease, such as ulceration and GI bleeding.

There is further increase in the risk of gastrointestinal adverse effects for celecoxib (gastrointestinal ulceration or other gastrointestinal complications), when celecoxib is taken concomitantly with acetylsalicylic acid (even at low doses).

The concomitant use of celecoxib and a non-aspirin NSAID should be avoided.

As the cardiovascular risks of celecoxib may increase with dose and duration of exposure, the shortest duration possible and the lowest effective daily dose should be used. The patient's need for symptomatic relief and response to therapy should be re-evaluated periodically, especially in patients with osteoarthritis (Recommended Dosage, Contraindications, Adverse Effects/ Undesirable Effects and Pharmacodynamics).

Patients with significant risk factors for cardiovascular events (e.g. hypertension, hyperlipidaemia, diabetes mellitus, smoking) should only be treated with celecoxib after careful consideration (see Pharmacodynamics).

COX-2 selective inhibitors are not a substitute for acetylsalicylic acid for prophylaxis of cardiovascular thrombo-embolic diseases because of their lack of antiplatelet effects. Therefore, antiplatelet therapies should not be discontinued.

As with other drugs known to inhibit prostaglandin synthesis, fluid retention and oedema have been observed in patients taking celecoxib. Therefore, celecoxib should be used with caution in patients with history of cardiac failure, left ventricular dysfunction or hypertension, and in patients with pre-existing oedema from any other reason, since prostaglandin inhibition may result in deterioration of renal function and fluid retention. Caution is also required in patients taking diuretic treatment or otherwise at risk of hypovolaemia.

As with all NSAIDs, celecoxib can lead to the onset of new hypertension or worsening of pre-existing hypertension, either of which may contribute to the increased incidence of cardiovascular events. Therefore, blood pressure should be monitored closely during the initiation of therapy with celecoxib and throughout the course of therapy.

Compromised renal or hepatic function and especially cardiac dysfunction are more likely in the elderly and therefore medically appropriate supervision should be maintained.

NSAIDs, including celecoxib, may cause renal toxicity. Patients at greatest risk for renal toxicity are those with impaired renal function, heart failure, liver dysfunction, those taking diuretics, ACE-inhibitors, angiotensin II receptor antagonists and the elderly (see Interactions with Other Medicaments). Such patients should be carefully monitored while receiving treatment with celecoxib.

Some cases of severe hepatic reactions, including fulminant hepatitis (some with fatal outcome), liver necrosis and, hepatic failure (some with fatal outcome or requiring liver transplant), have been reported with celecoxib. Among the cases that reported time to onset, most of the severe adverse hepatic events developed within one month after initiation of celecoxib treatment (see Adverse Effects/ Undesirable Effects).

If during treatment, patients deteriorate in any of the organ system functions described above, appropriate measures should be taken and discontinuation of celecoxib therapy should be considered.

Celecoxib inhibits CYP2D6. Although it is not a strong inhibitor of this enzyme, a dose reduction may be necessary for individually dose-titrated drugs that are metabolised by CYP2D6 (See Interaction with Other Medicaments).

Patients known to be CYP2C9 poor metabolisers should be treated with caution (see Pharmacokinetics).

Serious skin reactions, some of them fatal, including exfoliative dermatitis, Stevens-Johnson syndrome, and toxic epidermal necrolysis, have been reported very rarely in association with the use of celecoxib (see Adverse Effects/ Undesirable Effects). Patients appear to be at highest risk for these reactions early in the course of therapy: the onset of the reaction occurring in the majority of cases within the first month of treatment. Serious hypersensitivity reactions (including anaphylaxis, angioedema and drug rash with eosinophilia and systemic symptoms (DRESS, or hypersensitivity syndrome) have been reported in patients receiving celecoxib (see Adverse Effects/ Undesirable Effects). Patients with a history of sulphonamide allergy or any drug allergy may be at greater risk of serious skin reactions or hypersensitivity reactions (see Contraindications). Celecoxib should be discontinued at the first appearance of skin rash, mucosal lesions, or any other sign of hypersensitivity.

Celecoxib may mask fever and other signs of inflammation.

In patients on concurrent therapy with warfarin, serious bleeding events have occurred.

Caution should be exercised when combining celecoxib with warfarin and other oral anticoagulants (See Interactions with Other Medicaments).

Celecoxib capsules contain lactose. Patients with rare hereditary problems of galactose intolerance, the Lapp lactase deficiency or glucose-galactose malabsorption should not take this medicine.

Warning to prescriber when prescribing COX-2 Inhibitors to patients with risk factors of heart disease, hypertension (high blood pressure), hyperlipidemia, diabetes, smoking patient and patient with peripheral arterial disease.

Interactions with Other Medicaments

Pharmacodynamic interactions

Anticoagulant activity should be monitored particularly in the first few days after initiating or changing the dose of celecoxib in patients receiving warfarin or other anticoagulants since these patients have an increased risk of bleeding complications. Therefore, patients receiving oral anticoagulants should be closely monitored for their prothrombin time INR, particularly in the first few days when therapy with celecoxib is initiated or the dose of celecoxib is changed. Bleeding events in association with increases in prothrombin time have been reported, predominantly in the elderly, in patients receiving celecoxib concurrently with warfarin, some of them fatal.

NSAIDs may reduce the effect of diuretics and antihypertensive medicinal products. As for NSAIDs, the risk of acute renal insufficiency, which is usually reversible, may be increased in some patients with compromised renal function (e.g. dehydrated patients, patients on diuretics or elderly patients) when ACE inhibitors or angiotensin II receptor antagonists are combined with NSAIDs, including celecoxib. Therefore, the

combination should be administered with caution, especially in the elderly. Patients should be adequately hydrated and consideration should be given to monitoring of renal function after initiation of concomitant therapy, and periodically thereafter.

Coadministration of lisinopril with celecoxib can significantly cause being unresponsive to lisinopril in patients with hypertension.

Coadministration of NSAIDs and ciclosporin or tacrolimus have been suggested to increase the nephrotoxic effect of ciclosporin and tacrolimus. Renal function should be monitored when celecoxib and any of these drugs are combined.

Celecoxib can be used with low-dose acetylsalicylic acid but is not a substitute for acetylsalicylic acid for cardiovascular prophylaxis. As with other NSAIDs, an increased risk of gastrointestinal ulceration or other gastrointestinal complications compared to use of celecoxib alone was shown for concomitant administration of low-dose acetylsalicylic acid.

Pharmacokinetic interactions

Effects of celecoxib on other drugs

Celecoxib is an inhibitor of CYP2D6. During celecoxib treatment, the plasma concentrations of the CYP2D6 substrate dextromethorphan were increased by 136%. The plasma concentrations of drugs that are substrates of this enzyme may be increased when celecoxib is used concomitantly. Examples of drugs which are metabolised by CYP2D6 are antidepressants (tricyclics and SSRIs), neuroleptics, anti-arrhythmic drugs, etc. The dose of individually dose-titrated CYP2D6 substrates may need to be reduced when treatment with celecoxib is initiated or increased if treatment with celecoxib is terminated.

Celecoxib is able to inhibit CYP2C19 catalysed metabolism. Examples of drugs which are metabolised by CYP2C19 are diazepam, citalopram and imipramine.

Celecoxib had no clinically relevant effects on the pharmacokinetics of oral contraceptives (1 mg norethisterone /35 microg ethinylestradiol).

Celecoxib does not affect the pharmacokinetics of tolbutamide (CYP2C9 substrate), or glibenclamide to a clinically relevant extent.

In patients with rheumatoid arthritis celecoxib had no statistically significant effect on the pharmacokinetics (plasma or renal clearance) of methotrexate (in rheumatologic doses). However, adequate monitoring for methotrexate-related toxicity should be considered when combining these two drugs

In patients, co-administration of celecoxib 200 mg twice daily with 450 mg twice daily of lithium resulted in a mean increase in C_{max} of 16% and in AUC of 18% of lithium. Therefore, patients on lithium treatment should be closely monitored when celecoxib is introduced or withdrawn.

Effects of other drugs on celecoxib

In individuals who are CYP2C9 poor metabolisers and demonstrate increased systemic exposure to celecoxib, concomitant treatment with CYP2C9 inhibitors could result in further increases in celecoxib exposure. Such combinations should be avoided in known CYP2C9 poor metabolisers

Since celecoxib is predominantly metabolised by CYP2C9 it should be used at half the recommended dose in patients receiving fluconazole. Concomitant use of 200 mg single dose of celecoxib and 200 mg once daily of fluconazole, a potent CYP2C9 inhibitor, resulted in a mean increase in celecoxib C_{max} of 60% and in AUC of 130%. Concomitant use of inducers of CYP2C9 such as rifampicin, carbamazepine and barbiturates may reduce plasma concentrations of celecoxib.

Ketoconazole or antacids have not been observed to affect the pharmacokinetics of celecoxib.

Statement on Usage During Pregnancy and Lactation

Pregnancy

No clinical data on exposed pregnancies are available for celecoxib. The potential for human risk in pregnancy is unknown, but cannot be excluded. Celecoxib, as with other drugs inhibiting prostaglandin synthesis, may cause uterine inertia and premature closure of the ductus arteriosus during the last trimester. Celecoxib is contraindicated in pregnancy and in women who can become pregnant (see Contraindications and Warnings and Precautions). If a woman becomes pregnant during treatment, celecoxib should be discontinued.

Breastfeeding

Celecoxib is excreted in the milk of lactating rats at concentrations similar to those in plasma. Administration of celecoxib to a limited number of lactating women has shown a very low transfer of celecoxib into breast milk. Women who take celecoxib should not breastfeed.

Adverse Effects/ Undesirable Effects

Adverse reactions are listed by system organ class and ranked by frequency in **Table 1**

Adverse reactions

Adverse reactions reported as more than an isolated case are listed below, by system organ class and by frequency.

Within each frequency grouping, undesirable effects are presented in order of frequency, the most frequent first.

Table 1. Adverse Drug Reactions in Celecoxib Clinical Trials and Surveillance Experience
(MedDRA Preferred Terms)

Infections and infestations	
<i>Common:</i>	Sinusitis, upper respiratory tract infection, urinary tract infection
Blood and lymphatic system disorders	
<i>Uncommon:</i>	Anemia
<i>Rare:</i>	Leucopenia, thrombocytopenia
<i>Not known:</i>	Pancytopenia
Immune system disorders	
<i>Common:</i>	Allergy aggravated
<i>Not known:</i>	Serious allergic reactions, anaphylactic shock, anaphylaxis
Metabolism and nutrition disorders	
<i>Uncommon:</i>	Hyperkalemia
Psychiatric disorders	
<i>Common:</i>	Insomnia
<i>Uncommon:</i>	Anxiety, depression, tiredness
<i>Rare:</i>	Confusion
<i>Not known:</i>	Hallucinations
Nervous system disorders	
<i>Common:</i>	Dizziness, hypertonia
<i>Uncommon:</i>	Paraesthesia, somnolence, cerebral infarction
<i>Rare:</i>	Ataxia, taste alteration
<i>Not known:</i>	Headache, aggravated epilepsy, meningitis aseptic, ageusia, anosmia, fatal intracranial haemorrhage
Eye disorders	
<i>Uncommon:</i>	Blurred vision
<i>Not known:</i>	Conjunctivitis, ocular haemorrhage, retinal artery or vein occlusion
Ear and labyrinth disorders	
<i>Uncommon:</i>	Tinnitus, hypoacusis
Cardiac disorders	
<i>Common:</i>	Myocardial infarction
<i>Uncommon:</i>	Heart failure, palpitations, tachycardia

<i>Not known:</i>	Arrhythmia
Vascular disorders	
<i>Very common:</i>	Hyper-tension
<i>Uncommon:</i>	Hypertension aggravated
<i>Not known:</i>	Flushing, vasculitis, pulmonary embolism
Respiratory, thoracic, and mediastinal disorders	
<i>Common:</i>	Pharyngitis, rhinitis, cough, dyspnoea
<i>Not known:</i>	Bronchospasm
Gastrointestinal disorders	
<i>Common:</i>	Abdominal pain, diarrhoea, dyspepsia, flatulence, vomiting, dysphagia
<i>Uncommon:</i>	Constipation, eructation, gastritis, stomatitis, aggravation of gastrointestinal inflammation
<i>Rare:</i>	Duodenal, gastric, oesophageal, intestinal, and colonic ulceration; intestinal perforation; oesophagitis, melaena; pancreatitis
<i>Not known:</i>	Nausea, gastrointestinal haemorrhage, colitis/colitis aggravated
Hepatobiliary disorders	
<i>Uncommon:</i>	Abnormal hepatic function, increased SGOT and SGPT
<i>Rare:</i>	Elevation of hepatic enzymes
<i>Not known:</i>	Hepatic failure (sometimes fatal or requiring liver transplant), fulminant hepatitis (some with fatal outcome), liver necrosis, hepatitis jaundice
Skin and subcutaneous tissue disorders	
<i>Common:</i>	Rash, pruritus
<i>Uncommon:</i>	Urticaria, mouth pain and sores
<i>Rare:</i>	Alopecia, photosensitivity
<i>Not known:</i>	Ecchymosis, bullous eruption, exfoliative dermatitis, erythema multiforme, Stevens-Johnson syndrome, toxic epidermal necrolysis, drug rash with eosinophilia and systemic symptoms (DRESS) or hypersensitivity syndrome, angioedema, acute generalized exanthematous pustulosis
Musculoskeletal and connective tissue disorders	
<i>Uncommon:</i>	Leg cramps
<i>Not known:</i>	Arthralgia, myositis

Renal and urinary disorders	
<i>Uncommon:</i>	Increased creatinine, BUN increased
<i>Not known:</i>	Acute renal failure, interstitial nephritis, hyponatraemia
Reproductive system and breast disorders	
<i>Not known:</i>	Menstrual disorder NOS
General disorders and administrative site conditions	
<i>Common:</i>	Flu-like symptoms, peripheral oedema/ fluid retention
<i>Not known:</i>	Chest pain

Common: angina pectoris, irritable bowel syndrome, nephrolithiasis, blood creatinine increased, benign prostatic hyperplasia, weight increased.

Uncommon: helicobacter infection, herpes zoster, erysipelas, bronchopneumonia, labyrinthitis, gingival infection, lipoma, vitreous floaters, conjunctival haemorrhage, deep vein thrombosis, dysphonia, haemorrhoidal haemorrhage, frequent bowel movements, mouth ulceration, allergic dermatitis, ganglion, nocturia, vaginal haemorrhage, breast tenderness, lower limb fracture, blood sodium increased.

Overdose and Treatment

There is no clinical experience of overdose. In the event of suspected overdose, appropriate supportive medical care should be provided e.g. by eliminating the gastric contents, clinical supervision and, if necessary, the institution of symptomatic treatment. Dialysis is unlikely to be an efficient method of drug removal due to high protein binding.

Effects on Ability to Drive and Use Machine

Patients who experience dizziness, vertigo or somnolence while taking celecoxib should refrain from driving or operating machinery.

Storage Conditions

Store below 30°C

Dosage Forms and Packaging Available

- The product is available in the paper box containing 10 and 100 capsules.
- 10 capsules are packed in a blister pack (PVC/ALU) and then be closed in a secondary paper box containing 1 and 10 blister(s).

Name and Address of Manufacturer

MacroPhar Co., Ltd.

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Name and Address of Marketing Authorization Holder

Stadpharm Sdn. Bhd.

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Date of Revision of Package Insert

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